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United States Patent Application Publication	20250271733
Kind Code	A1
Publication Date	August 28, 2025
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OPTICAL APPARATUS

Abstract

An optical apparatus includes a first optical member and a first holding member configured to hold the first optical member and a second optical member located on an inner side of the first optical member that are sealed with a first sealing member A space between the first optical member and the second optical member is sealed.

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Family ID:	1000008491140
Appl. No.:	19/061205
Filed:	February 24, 2025

Foreign Application Priority Data

JP	2024-026140	Feb. 26, 2024
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Publication Classification

Int. Cl.: G03B17/08 (20210101); G02B1/04 (20060101); G02B7/02 (20210101); G02B23/00 (20060101); G03B17/12 (20210101)

U.S. Cl.:

CPC G03B17/08 (20130101); G02B1/041 (20130101); G02B7/021 (20130101); G02B7/025 (20130101); G02B7/028 (20130101); G02B23/145 (20130101); G03B17/12 (20130101);

Background/Summary

BACKGROUND

Technical Field

[0001] The present disclosure relates to a sealing structure of an optical apparatus having a plurality of optical members arrayed in an optical axis direction. For example, the present disclosure relates to a sealing structure of an optical apparatus which constitutes a finder unit of a camera main body.

Description of the Related Art

[0002] Conventionally, a difference has occurred in the ambient temperature and a temperature of an inner side of a camera when the camera is suddenly relocated to a cold environment such as an environment of winter, a cold weather, or a night time, from a warm environment such as a room temperature environment.

[0003] Because a temperature is lowered on the inner side of an eyepiece lens which is in contact with the ambient air, a saturated moisture amount of moisture inside the camera is decreased. As a result, dew condensation may occur on a surface on the inner side of the eyepiece lens which is in contact with the ambient air.

[0004] Further, there is a case where the camera is turned on under a high-humidity and low-temperature environment such as an environment of a rainy weather.

[0005] When the camera is turned on, a temperature of an inner side of the camera is increased due to heat generated from an integrated circuit mounted on an electronic substrate or heat generated from a backlight used for a display element included in a viewfinder. Accordingly, dew condensation may similarly occur on the surface on the inner side of the eyepiece lens which is in contact with the ambient air.

[0006] In any of the above-described cases, there is a disadvantage that visibility is degraded by dew condensation because the user cannot wipe the inner side of the eyepiece lens.

[0007] With respect to the above-described disadvantage, dew condensation can be reduced by a structure which prevents entry of moisture into the inner side of the eyepiece lens in contact with the ambient air.

[0008] Japanese Patent Application Laid-Open No. 2021-135354 discusses a method which suppresses entry of moisture into the inner side of the eyepiece lens in contact with the ambient air, by sealing a space between an outermost lens and a second-outermost lens by annularly bonding the outermost lens and the second-outermost lens with an adhesive medium.

SUMMARY

[0009] An aspect of the present disclosure is to provide a holding structure of optical members, capable of reducing worsening of surface accuracy, degradation of resolution, and distortion of an image, even when either an eyepiece lens in contact with the ambient air or a second-outermost eyepiece lens is made of a plastic material.

[0010] Another aspect of the present disclosure is to provide a holding structure of optical members, capable of suppressing dew condensation caused by entry of moisture into the inner side of the eyepiece lens.

[0011] According to an aspect of the present disclosure, an optical apparatus includes a first optical member and a second optical member located on an inner side of the first optical member from among an optical system including at least two optical members, a first holding member configured to hold the first optical member and the second optical member, a first sealing member configured to seal a gap between the first optical member and the first holding member, and a second sealing member configured to seal a gap between the first holding member and the second optical member, wherein the second optical member has an abutting portion on an outer circumferential portion, at which the second sealing member is in close contact with the second optical member, wherein the first holding member has an abutting portion on an outer circumferential portion, at which the second sealing member is in close contact with the first holding member, wherein the first optical

member and the first holding member are sealed with the first sealing member, and wherein a space between the first optical member and the second optical member is sealed.

[0012] Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] FIGS. 1A and 1B are external perspective views of a digital camera.

[0014] FIG. 2 is a cross-sectional view of an eyepiece unit.

[0015] FIG. 3 is an exploded perspective view of the eyepiece unit.

[0016] FIG. 4 is an exploded perspective view illustrating a state before an optical member is mounted on a lens holding member.

[0017] FIG. 5 is an exploded perspective view illustrating a state before a spacer is mounted on the lens holding member.

[0018] FIG. 6 is an exploded perspective view illustrating a state before an optical member is mounted on the lens holding member.

[0019] FIG. 7 is an exploded perspective view illustrating a state before a spacer is mounted on the lens holding member.

[0020] FIG. 8 is an exploded perspective view illustrating a state before an elastic member is mounted on the lens holding member.

[0021] FIG. 9 is an exploded perspective view illustrating a state before an optical member is fixed to a lens holding member with a sealing member.

[0022] FIG. 10 is an exploded perspective view illustrating a state before an optical member is mounted on the lens holding member.

[0023] FIG. 11 is an exploded perspective view illustrating a state before a gap between the lens holding member and the optical member is sealed with a sealing member.

[0024] FIG. 12 is an exploded perspective view illustrating a state before the lens holding members are combined.

DESCRIPTION OF THE EMBODIMENTS

[0025] By the conventional technique discussed in Japanese Patent Application Laid-Open No. 2021-135354, entire circumferences of lenses can be bonded when an eyepiece lens in contact with the ambient air and a second-outermost lens are made of glass because the lenses are not affected by shrinkage of an adhesive agent that occurs when the adhesive agent is hardened.

[0026] However, it is often the case that a plastic material is used for an eyepiece lens in order to reduce cost and a weight.

[0027] In a case where the entire circumferences of the lenses are bonded to each other when a plastic material is used for an eyepiece lens, the lens is deformed by shrinkage of an adhesive agent that occurs when the adhesive agent is hardened, so that surface accuracy of the lens surface is worsened. This causes degradation of resolution and distortion of an image.

[0028] Therefore, when a lens made of a plastic material is to be bonded to another lens, it is often the case that the lenses are partially bonded to each other instead of being bonded in the entire circumferences, in order to prevent worsening of surface accuracy caused by deformation of the lens.

[0029] In other words, the entire circumferences of the lenses cannot be adhered to each other in a case where at least either one of the eyepiece lens in contact with the ambient air and the second-outermost lens is made of plastic materials.

[0030] Therefore, it is not possible to prevent entry of moisture into the inner side of the eyepiece lens by the conventional technique discussed in Japanese Patent Application Laid-Open No. 2021-

135354.

[0031] Hereinafter, exemplary embodiments of the present disclosure will be described in detail with reference to the appended drawings.

(External Perspective View of Digital Camera **100**)

[0032] FIGS. **1A** and **1B** are external perspective views of a digital camera **100** as one example of the optical apparatuses to which the present disclosure is applicable.

[0033] A mobile phone, a game machine, a tablet terminal, a personal computer, a watch-type or an eyewear-type information terminal, and a head-mounted display are given as the examples of the optical apparatuses.

[0034] The present disclosure is also applicable to products having eyepiece units (e.g., a monocular viewer, a binocular viewer, and a range-finding meter), and eyepiece attachments attached to the eyepiece units (e.g., a rain cover, a magnifier, and an angle finder).

[0035] FIG. **1A** illustrates an external perspective view from a front face of the digital camera **100**. FIG. **1B** illustrates an external perspective view from a back face of the digital camera **100**.

[0036] In FIGS. **1A** and **1B**, a portion of an eyecup **802** which comes into contact with the user's eye is made of a soft material in order to prevent light from entering an interior of a viewfinder from a gap between an eyepiece frame **801** and the user's eye.

[0037] A position of an electronic viewfinder (EVF) module **29** (illustrated in FIG. **2**) is adjusted by a diopter adjustment dial **900** based on the diopter of the user.

[0038] An eyepiece unit **16** is a magnifying optical system of an eyepiece viewfinder (i.e., look-into type viewfinder). The user can magnify and visually recognize, via the eyepiece unit **16**, an image displayed on the EVF module **29** arranged inside.

[0039] The EVF module **29** is a display apparatus such as a liquid crystal display apparatus or an organic electroluminescence display apparatus, which displays an image based on an analog signal received from a digital-to-analog (D/A) conversion unit (not illustrated).

[0040] When the digital camera **100** is held at a normal position, a display of the EVF module **29** in a vertical direction is shorter than a display in a horizontal direction.

(Configuration of Eyepiece Unit)

[0041] Next, a configuration of the eyepiece unit **16** as an optical apparatus according to the present exemplary embodiment will be described with reference to FIGS. **2** and **3**.

[0042] FIG. **2** is a cross-sectional view of a configuration of the eyepiece unit **16**.

[0043] FIG. **3** is an exploded perspective view of the eyepiece unit **16**.

[0044] FIG. **2** illustrates a vertical cross-sectional view of the eyepiece unit **16** having an optical axis **1000** at a center, when the digital camera **100** (not illustrated) is held in a normal position. In FIG. **2**, a left side is a side of the EVF module **29**, and a right side is a side of an observer.

[0045] The eyepiece unit **16** is a magnifying optical system serving as a viewfinder mounted on a digital camera, a mirrorless camera, or a digital video camera, and that magnifies an image displayed on the EVF module **29** for the observer to observe. The eyepiece unit **16** is integrally or removably mounted on an image capturing apparatus.

[0046] Optical members **402**, **408**, and **413** are plastic lenses, and an optical member **401** is a flat plate-like cover window. The optical member **401** is made of plastic or glass.

[0047] The optical member **402** has a convex lens shape projecting to a side of the optical member **401**.

[0048] Although the optical member **401** described in the present exemplary embodiment is a flat plate-like member, the optical member **401** may be a member having a lens shape.

[0049] In the present exemplary embodiment, because the optical member **402** projects to a side of the optical member **401** having a flat plate shape, the convex lens (i.e., optical member **402**) enters a space between the optical members **401** and **402** when the space is sealed as described below. As a result, it is possible to reduce a volume.

[0050] Further, a distance between the optical members **401** and **402** in a vicinity of the optical axis

1000 can be reduced, so that a layer of air in the vicinity of the optical axis **1000** becomes thin. Consequently, a central portion of the optical member **401** through which the user looks into the digital camera **100** is less likely to be fogged when the digital camera **100** is used.

[0051] Further, because the optical member **401** which is in contact with the ambient air is a flat plate-like member, variation in temperatures is less likely to occur in the optical member **401**. Therefore, the optical member **401** is uniformly less likely to be fogged.

[0052] A lens holding member **403** holds the optical members **401** and **402**, and a lens holding member **409** holds the optical members **408** and **413**.

[0053] A distance between the optical members **413**, **408**, and **402** is determined by spacers **411** and **412**. The spacers **411** and **412** may also function as masks for preventing unnecessary light from entering the optical members **413**, **408**, and **402**.

[0054] An elastic member **410** presses the optical members **413**, **408**, and **402** and the spacers **412** and **411** against the lens holding members **409** and **403**. The elastic member **410** is formed of a material having elasticity, e.g., silicon rubber.

[0055] A sealing member **404** seals the optical member **401** and the lens holding member **403**, so that the optical member **401** is held by the lens holding member **403**. In the present exemplary embodiment, the sealing member **404** is a double-sided adhesive tape consisting of a polyolefin foamed base material on which an acrylic adhesive material is applied.

[0056] In a case where the optical member **401** is made of glass, the optical member **401** has resistance to an external force which causes deformation.

[0057] In this case, an ultraviolet (UV) curable adhesive agent or an anaerobic adhesive agent can be used as the sealing member **404** instead of the double-sided adhesive tape, as long as a gap between the lens holding member **403** and the optical member **401** can be sealed thereby.

[0058] Further, the sealing member **404** may also function as a mask for preventing unnecessary light from entering the optical members **402** and **401**.

[0059] A sealing member **405** is a sheet-like sealing member for sealing the optical member **402** and the lens holding member **403**.

[0060] According to the present exemplary embodiment, the sealing member **405** can be a one-sided adhesive tape or a double-sided adhesive tape consisting of a polyolefin foamed base material on which an acrylic adhesive material is applied, or can be a non-adhesive elastic sheet made of a material such as silicon rubber.

[0061] Further, the sealing member **405** may also function as a mask for preventing unnecessary light from entering the optical members **402** and **408**. Further, the effect of the sealing members **404** and **405** can be improved by using a material such as a polyethylene film having low moisture permeability as the base materials.

[0062] The EVF module **29** can be moved in a direction parallel to the optical axis **1000** by rotating the diopter adjustment dial **900** (illustrated in FIG. 1B) through a cam mechanism (not illustrated).

[0063] In this way, the user can adjust a position of the EVF module **29** based on the diopter of the user.

[0064] In the present exemplary embodiment, the EVF module **29** is moved by the diopter adjustment dial **900**. Therefore, even when the user adjusts the diopter, relative positions of the optical members **401** and **402** are not changed.

[0065] In accordance with the display of the EVF module **29**, an optical effective range in the up-down direction of each of the optical members **401**, **402**, **408**, and **412** is shorter than an optical effective range in the right-left direction when the digital camera **100** is held at a normal position.

[0066] Next, with reference to FIGS. 4 to 12, a configuration of the eyepiece unit **16** is described in detail according to the assembly order.

(Exploded Perspective View of a State Before Optical Member **413** is Mounted on Lens Holding Member **409**)

[0067] FIG. 4 is an exploded perspective view of a state before the optical member **413** is mounted

on the lens holding member **409**.

[0068] A position of the optical member **413** in a radial direction with respect to the optical axis **1000** (illustrated in FIG. 2) of the lens is determined by radial bearing faces **413c1** to **413c6** provided on an outer circumference of the optical member **413**.

[0069] The radial bearing faces **413c1** to **413c6** respectively abut on radial bearing faces **409c1** to **409c6** provided on an inner circumference of the lens holding member **409**, so that a position of the optical member **413** in the radial direction, at which the optical axis of the optical member **413** coincides with the optical axis **1000** of the eyepiece unit **16**, is determined.

[0070] A position of the optical member **413** in a thrust direction with respect to the optical axis **1000** of the eyepiece unit **16** is determined by thrust bearing faces **413a1** to **413a3** and **413b1** to **413b3**.

[0071] The above-described thrust bearing faces **413a1** to **413a3** and **413b1** to **413b3** are provided on the outer circumferential portion of the optical member **413**, on the outer side of the optical effective range of the optical member **413**. The thrust bearing faces **413a1** to **413a3** and **413b1** to **413b3** are respectively provided at three points on the front side and three points on the back side of the optical member **413**, and positions thereof on the front side and positions thereof on the back side are substantially the same on a projection face viewed in the optical axis direction.

[0072] The optical member **413** is held by making only the thrust bearing faces **413a1** to **413a3** and **413b1** to **413b3** contact with the lens holding member **409**. With this configuration, bending stress is less likely to be placed on the optical member **413**, so that degradation of optical performance caused by deformation is less likely to occur.

[0073] The thrust bearing faces **413a1** to **413a3** respectively abut on thrust bearing faces **409a1** to **409a3** provided on the lens holding member **409**, so that the position of the optical member **413** in the thrust direction is determined.

(Exploded Perspective View of a State Before Spacer **412** is Mounted on Lens Holding Member **409**)

[0074] FIG. 5 is an exploded perspective view of a state before the spacer **412** is mounted on the lens holding member **409** after the optical member **413** is mounted thereon.

[0075] A position of the spacer **412** in the radial direction with respect to the optical axis **1000** (illustrated in FIG. 2) of the lenses is determined by radial bearing faces **412c1** to **412c6** provided on an outer circumference of the spacer **412**.

[0076] The radial bearing faces **412c1** to **412c6** respectively abut on radial bearing faces **409d1** to **409d6** provided on an inner circumference of the lens holding member **409**, so that a position of the spacer **412** in the radial direction, at which the optical axis of the spacer **412** coincides with the optical axis **1000** of the eyepiece unit **16**, is determined.

[0077] A position of the spacer **412** in the thrust direction with respect to the optical axis **1000** of the eyepiece unit **16** is determined by thrust bearing faces **412a1** to **412a3** and **412b1** to **412b3**.

[0078] The thrust bearing faces **412a1** to **412a3** and **412b1** to **412b3** are respectively arranged at three points on the front side and three points on the back side of the spacer **412**, and positions thereof on the front side and positions thereof on the back side are substantially the same on a projection face viewed in the optical axis direction.

[0079] Because the spacer **412** is held by making only the thrust bearing faces **412a1** to **412a3** and **412b1** to **412b3** contact with the lens holding member **409**, bending stress is less likely to be placed on the spacer **412**.

[0080] With this configuration, positional accuracy of the thrust bearing faces **412a1** to **412a3** and **412b1** to **412b3** is less likely to be degraded.

[0081] The thrust bearing faces **412a1** to **412a3** respectively abut on the thrust bearing faces **413b1** to **413b3** provided on the optical member **413**, so that a position of the spacer **412** in the thrust direction is determined.

(Exploded Perspective View of a State Before Optical Member **408** is Mounted on Lens Holding

Member **409**)

[0082] FIG. **6** is an exploded perspective view of a state before the optical member **408** is mounted on the lens holding member **409** after the optical member **413** and the spacer **412** are mounted thereon.

[0083] A position of the optical member **408** in the radial direction with respect to the optical axis **1000** (illustrated in FIG. **2**) of the lens is determined by radial bearing faces **408c1** to **408c6** provided on an outer circumference of the optical member **408**.

[0084] The radial bearing faces **408c1** to **408c6** respectively abut on radial bearing faces **409f1** to **409f6** provided on an inner circumference of the lens holding member **409**, so that a position of the optical member **408** in the radial direction, at which the optical axis of the optical member **408** coincides with the optical axis **1000** of the eyepiece unit **16**, is determined.

[0085] A position of the optical member **408** in the thrust direction with respect to the optical axis **1000** of the eyepiece unit **16** is determined by thrust bearing faces **408a1** to **408a3** and **408b1** to **408b3**.

[0086] The above-described thrust bearing faces **408a1** to **408a3** and **408b1** to **408b3** are provided on the outer circumferential portion of the optical member **408**, on the outer side of the optical effective range of the optical member **408**. The thrust bearing faces **408a1** to **408a3** and **408b1** to **408b3** are respectively provided at three points on the front side and three points on the back side of the optical member **408**, and positions thereof on the front side and positions thereof on the back side are substantially the same on a projection face viewed in the optical axis direction.

[0087] The optical member **408** is held by making only the thrust bearing faces **408a1** to **408a3** and **408b1** to **408b3** contact with the lens holding member **409**. With this configuration, bending stress is less likely to be placed on the optical member **408**, so that degradation of optical performance caused by deformation is less likely to occur.

[0088] The thrust bearing faces **408a1** to **408a3** respectively abut on thrust bearing faces **409e1** to **409e3** provided on the lens holding member **409**, so that a position of the optical member **408** in the thrust direction is determined.

[0089] As described above, the optical members **413** and **408** are respectively in contact with and held by the lens holding member **409** at different positions.

[0090] With this configuration, in a case where inclination occurs in the optical members **413** and **408** depending on a dimension of a finished component, each of the optical members **413** and **408** can individually be corrected by modifying the heights of the thrust bearing faces at three positions. (Exploded Perspective View of a State Before Spacer **411** is Mounted on Lens Holding Member **409**)

[0091] FIG. **7** is an exploded perspective view of a state before the spacer **411** is mounted on the lens holding member **409** after the optical member **413**, the spacer **412**, and the optical member **408** are mounted thereon.

[0092] A position of the spacer **411** in the radial direction with respect to the optical axis **1000** (illustrated in FIG. **2**) of the lenses is determined by radial bearing faces **411c1** to **411c6** provided on an outer circumference of the spacer **411**.

[0093] The radial bearing faces **411c1** to **411c6** respectively abut on radial bearing faces **409g1** to **409g6** provided on an inner circumference of the lens holding member **409**, so that a position of the spacer **411** in the radial direction, at which the optical axis of the spacer **411** coincides with the optical axis **1000** of the eyepiece unit **16**, is determined.

[0094] A position of the spacer **411** in the thrust direction with respect to the optical axis **1000** of the eyepiece unit **16** is determined by thrust bearing faces **411a1** to **411a3** and **411b1** to **411b3**.

[0095] The thrust bearing faces **411a1** to **411a3** and **411b1** to **411b3** are respectively provided at three points on the front side and three points on the back side of the spacer **411**, and positions thereof on the front side and positions thereof on the back side are substantially the same on a projection face viewed in the optical axis direction.

[0096] Because the spacer **411** is held by making only the thrust bearing faces **411a1** to **411a3** and **411b1** to **411b3** contact with the lens holding member **409**, bending stress is less likely to be placed on the spacer **411**.

[0097] With this configuration, positional accuracy of the thrust bearing faces **411a1** to **411a3** and **411b1** to **411b3** is less likely to be degraded.

[0098] The thrust bearing faces **411a1** to **411a3** respectively abut on the thrust bearing faces **408b1** to **408b3** provided on the optical member **408**, so that a position of the spacer **411** in the thrust direction is determined.

(Exploded Perspective View of a State Before Elastic Member **410** is Mounted on Lens Holding Member **409**)

[0099] FIG. **8** is an exploded perspective view of a state before an elastic member **410** is mounted on the lens holding member **409** after the optical member **413**, the spacer **412**, the optical member **408**, and the spacer **411** are mounted thereon.

[0100] Thrust bearing faces **410a1** to **410a3** and **410b1** to **410b3** are provided on the elastic member **410**.

[0101] The thrust bearing faces **410a1** to **410a3** respectively abut on the thrust bearing faces **412b1** to **412b3** provided on the spacer **412**, and the thrust bearing faces **410b1** to **410b3** respectively abut on the thrust bearing faces **411b1** to **411b3** provided on the spacer **411**.

[0102] Further, as described below, the sealing member **405** abuts on the elastic member **410** to compress the elastic member **410**.

[0103] Therefore, elastic force is generated to cause the optical member **413** to be urged against the lens holding member **409** via the spacer **412**, and to cause the optical member **408** to be urged against the lens holding member **409** via the spacer **411**.

[0104] Then, the optical member **402** is urged against the lens holding member **403** via the sealing member **405**, so that the thrust positions of the optical members **413** and **408** are stabilized.

(Exploded Perspective View of a State Before Optical Member **401** is Fixed with Sealing Member **404**)

[0105] FIG. **9** is an exploded perspective view of a state before the optical member **401** is fixed to the lens holding member **403** by the sealing member **404**.

[0106] The sealing member **404** is a double-sided adhesive tape with adhesive materials applied on both sides. The sealing member **404** is held between the lens holding member **403** and the optical member **401** to seal a gap between an opening portion **403d** of the lens holding member **403** and the optical member **401**. In the present exemplary embodiment, a double-sided adhesive tape is used as the sealing member **404**.

[0107] However, in a case where the optical member **401** is a lens made of glass, image distortion caused by shrinkage of an adhesive agent does not occur because optical performance thereof is not affected by shrinkage of the adhesive agent. Therefore, in this case, the gap may be sealed by applying the adhesive agent to the entire circumference of the optical member **401**.

(Exploded Perspective View of a State Before Optical Member **402** is Mounted on Lens Holding Member **403**)

[0108] FIG. **10** is an exploded perspective view of a state before the optical member **402** is mounted on the lens holding member **403** after the optical member **401** is sealed and fixed to the lens holding member **403**.

[0109] A position of the optical member **402** in the radial direction with respect to the optical axis **1000** (illustrated in FIG. **2**) of the lenses is determined by radial bearing faces **402b1** to **402b6** provided on an outer circumference of the optical member **402**.

[0110] The radial bearing faces **402b1** to **402b6** respectively abut on radial bearing faces **403b1** to **403b6** provided on an inner circumference of the lens holding member **403**, so that a position of the optical member **402** in the radial direction, at which the optical axis of the optical member **402** coincides with the optical axis **1000** of the eyepiece unit **16**, is determined.

[0111] A position of the optical member **402** in the thrust direction with respect to the optical axis **1000** of the eyepiece unit **16** is determined by thrust bearing faces **402a1** to **402a3**.

[0112] The thrust bearing faces **402a1** to **402a3** respectively abut on thrust bearing faces **403a1** to **403a3** provided on the lens holding member **403**, so that a position of the optical member **402** in the thrust direction is determined.

(Exploded Perspective View of a State Before Gap is Sealed with Sealing Member **405**)

[0113] FIG. **11** is an exploded perspective view of a state before a gap between the lens holding member **403** and the optical member **402** is sealed with the sheet-like sealing member **405**.

[0114] A planar face portion **402c** vertical to the optical axis **1000** is provided on the optical member **402** on the outer circumference portion, on the outer side of the optical effective range of the optical member **402**.

[0115] Further, a planar face portion **403c** vertical to the optical axis **1000** is provided on the lens holding member **403**, at a portion which is to be located on the outer circumference portion of the optical member **402** when the optical member **402** is mounted on the lens holding member **403**.

[0116] When the optical member **402** is mounted on the lens holding member **403**, the above-described two planar face portions **402c** and **403c** have substantially the same heights.

[0117] An adhesive agent is applied to one of the surfaces of the sealing member **405** facing the optical member **402**.

[0118] The sealing member **405** is applied to both of the planar face portion **403c** of the lens holding member **403** and the planar face portion **402c** of the optical member **402** having substantially the same heights without any gap, so that the lens holding member **403** and the optical member **402** are sealed.

[0119] Through the above-described configuration, it is possible to seal the plastic optical member **402** susceptible to deformation by an external force without placing a load that may cause deformation.

[0120] As described above, only the optical member **401**, which is in contact with the ambient air, and the second-outermost optical member **402** are mounted on the lens holding member **403** different from the lens holding member **409**.

[0121] In this way, when the optical member **402** is mounted on the lens holding member **403**, the sheet-like sealing member **405** can easily be applied to the planar face portions **402c** and **403c** having substantially the same heights.

(Exploded Perspective View of a State Before Lens Holding Member **403** is Combined with Lens Holding Member **409** after Optical Members **401** and **402** are Sealed and Fixed to the Lens Holding Member **403**)

[0122] FIG. **12** is an exploded perspective view of a state before the lens holding member **403**, on which the optical members **401** and **402** are sealed and fixed, is combined with the lens holding member **409**, on which the optical member **413**, the spacer **412**, the optical member **408**, the spacer **411**, and the elastic member **410** are mounted.

[0123] Positioning pins **409h** provided on the lens holding member **409** fit into positioning holes **403d** provided in the lens holding member **403**, so that a position of the lens holding member **403** in the radial direction is determined.

[0124] Further, planar face portions **403e** (illustrated in FIG. **11**) of the lens holding member **403** respectively abut on planar face portions **409j** of the lens holding member **409**, so that a position of the lens holding member **403** in the thrust direction is determined. Then, the two lens holding members **409** and **403** are fixed with screws **414**.

[0125] At this time, the sealing member **405** abuts on the elastic member **410** to compress the elastic member **410**, so that elastic force is generated.

[0126] As a result, the optical member **413** is urged against the lens holding member **409** via the spacer **412**, and the optical member **408** is urged against the lens holding member **409** via the spacer **411**.

[0127] The optical member **402** is urged against the lens holding member **403** via the sealing member **405**, so that the positions of the optical members **413** and **408** in the thrust direction are stabilized.

[0128] Further, because the sealing member **405** is urged against both of the optical member **402** and the lens holding member **403** by the elastic member **410**, the optical member **402** and the lens holding member **403** are sealed more tightly.

[0129] In the present exemplary embodiment, the optical members **413** and **408** are urged against the lens holding member **403** by the elastic member **410** via the spacers **411** and **412**. However, the optical members **413** and **408** may be urged against the lens holding member **403** by the elastic member **410** which is directly in contact with the optical member **413** or **408**.

[0130] Through the above-described configuration, a space between the optical members **401** and **402** is sealed with the sealing member **404**, the lens holding member **403**, and the sheet-like sealing member **405**. With this configuration, it is possible to prevent entry of moisture from the outside.

[0131] Accordingly, a humidity is less likely to be high on an inner face side of the optical member **401**, so that an amount of moisture in a sealed space is less likely to exceed the saturated moisture amount even when a temperature on the inner face side of the optical member **401** is decreased due to the influence of the ambient air. Therefore, it is possible to prevent the optical member **401** from being fogged easily.

[0132] Further, only the space between the optical members **401** and **402** is sealed.

[0133] With this configuration, an absolute amount of moisture present in the sealed space can be reduced.

[0134] Therefore, even if the temperature on the inner face side of the optical member **401** decreases due to the influenced of the ambient air, and the amount of moisture in the sealed space exceeds the saturated moisture amount, it is possible to suppress fogging to a minimum extent.

[0135] Further, because the optical member **402** projects to a side of the optical member **401** having a flat plate shape, the convex lens (i.e., optical member **402**) enters a space between the optical members **401** and **402** when the space is sealed. Therefore, it is possible to reduce the volume.

[0136] Further, a distance L (illustrated in FIG. 2) between the optical members **401** and **402** in a vicinity of the optical axis **1000** is made shorter than a thickness t (illustrated in FIG. 2) of the optical member **401**, so that a layer of air in the vicinity of the optical axis **1000** becomes thin. Consequently, a central portion of the optical member **401** through which the user looks into the digital camera **100** is less likely to be fogged when the digital camera **100** is used.

[0137] Further, the optical member **401** which is in contact with the ambient air is a flat plate-like member. With this configuration, variation in temperatures is less likely to occur in the optical member **401**, so that the optical member **401** is uniformly less likely to be fogged.

[0138] Further, even when at least either one of the optical member **401** in contact with the ambient air and the second-outermost optical member **402** is made of a plastic material, the space can be sealed with the sheet-like sealing member **405** without placing a load on the lenses (i.e., optical members **401** and **402**).

[0139] Therefore, it is possible to prevent dew condensation without causing worsening of surface accuracy, degradation of resolution, and distortion of an image.

[0140] The characteristics of the present exemplary embodiment will be described below in an organized manner.

[0141] The first characteristic will be described with reference to FIG. 2.

[0142] The optical apparatus includes a first optical member **401** in contact with ambient air and a second optical member **402** located on an inner side of the first optical member **401** from among an optical system including at least two optical members.

[0143] The optical apparatus further includes a first holding member **403** capable of holding the first optical member **401** and the second optical member **402** and a first sealing member **404** for

sealing a gap between the first optical member **401** and the first holding member **403**.

[0144] The optical apparatus further includes a second sealing member **405** for sealing a gap between the first holding member **403** and the second optical member **402**.

[0145] The second optical member **402** has an abutting portion **402c** on an outer circumferential portion, at which the second sealing member **405** is in contact with the second optical member **402**, and the first holding member **403** has an abutting portion **403c** on an outer circumferential portion, at which the second sealing member **405** is in close contact with the first holding member **403**.

[0146] The first optical member **401** and the first holding member **403** are sealed with the first sealing member **404**, and a space between the first optical member **401** and the second optical member **402** is sealed.

[0147] The second characteristic will be described with reference to FIGS. 2 and 3.

[0148] The first optical member **401** is a flat plate.

[0149] The third characteristic will be described with reference to FIGS. 2 and 3.

[0150] The second optical member **402** is made of plastic.

[0151] The fourth characteristic will be described with reference to FIG. 2.

[0152] The second optical member **402** is a convex shape lens convexed toward the first optical member **401**.

[0153] The fifth characteristic will be described with reference to FIG. 2.

[0154] A distance between the first optical member **401** and the second optical member **402** on an optical axis is shorter than a thickness of the first optical member **401**.

[0155] The sixth characteristic will be described with reference to FIG. 2.

[0156] The first sealing member **404** is a double-sided adhesive tape.

[0157] The seventh characteristic will be described with reference to FIG. 2.

[0158] The first sealing member **404** is an adhesive agent.

[0159] The eighth characteristic will be described with reference to FIGS. 2 and 9.

[0160] Either one of the first sealing member **404** and the second sealing member **405** also functions as a mask for limiting an optical path.

[0161] The ninth characteristic will be described with reference to FIG. 3.

[0162] The optical apparatus further includes an electronic display apparatus **29**, and the first optical member **401** and the second optical member **402** constitute a magnifying optical system through which the electronic display apparatus **29** is magnified and observed.

[0163] The tenth characteristic will be described with reference to FIG. 2.

[0164] An optical effective range of each of the first optical member **401** and the second optical member **402** is shorter in a vertical direction than in a horizontal direction.

[0165] The eleventh characteristic will be described with reference to FIGS. 1A and 1B.

[0166] The optical apparatus further includes a diopter adjustment unit **900** for adjusting diopter of the electronic display apparatus **29**, wherein the diopter can be adjusted by the diopter adjustment unit **900** by moving a position of the electronic display apparatus **29** in an optical axis direction.

[0167] The twelfth characteristic will be described with reference to FIGS. 1A and 1B.

[0168] A relative positional relationship between the first optical member **401** and the second optical member **402** does not change when the diopter is adjusted by the diopter adjustment unit **900**.

[0169] The thirteenth characteristic will be described with reference to FIG. 2.

[0170] The optical apparatus further includes a third optical member **408** arranged on an inner side of the second optical member **402**, a second holding member **409** for holding the third optical member **408**, and an urging member **410** for urging the first holding member **403** against the second holding member **409**.

[0171] The second sealing member **405** is in contact with the urging member **410** to compress the urging member **410**, thereby urging the first holding member **403** against the second holding member **409** and determining a position in the optical axis direction.

[0172] The fourteenth characteristic will be described with reference to FIG. 2.

[0173] The abutting portion of the second optical member **402** is a planar face portion **402c** at which the second sealing member **405** is in close contact with the second optical member **402**.

[0174] The abutting portion of the first holding member **403** is a planar face portion **403c** at which the second sealing member **405** is in close contact with the first holding member **403**.

[0175] Next, combinations of the exemplary embodiments included in the present disclosure will be described.

(Configuration 1)

[0176] An optical apparatus includes a first optical member **401** and a second optical member **402** located on an inner side of the first optical member **401** from among an optical system including at least two optical members, a first holding member **403** configured to hold the first optical member **401** and the second optical member **402**, a first sealing member **404** for sealing a gap between the first optical member **401** and the first holding member **403**, and a second sealing member **405** for sealing a gap between the first holding member **403** and the second optical member **402**, wherein the second optical member **402** has an abutting portion **402c** on an outer circumferential portion, at which the second sealing member **405** is in close contact with the second optical member **402**, wherein the first holding member **403** has an abutting portion **403c** on an outer circumferential portion, at which the second sealing member **405** is in close contact with the first holding member **403**, wherein the first optical member **401** and the first holding member **403** are sealed with the first sealing member **404**, and wherein a space between the first optical member **401** and the second optical member **402** is sealed.

(Configuration 2)

[0177] The optical apparatus according to Configuration 1, wherein the first optical member **401** is a flat plate.

(Configuration 3)

[0178] The optical apparatus according to Configuration 1 or 2, wherein the second optical member **402** is made of plastic.

(Configuration 4)

[0179] The optical apparatus according to any one of Configurations 1 to 3, wherein the second optical member **402** is a convex shape lens convexed toward the first optical member **401**.

(Configuration 5)

[0180] The optical apparatus according to any one of Configurations 2 to 4, wherein a distance between the first optical member **401** and the second optical member **402** on an optical axis is shorter than a thickness of the first optical member **401**.

(Configuration 6)

[0181] The optical apparatus according to any one of Configurations 1 to 5, wherein the first sealing member **404** is a double-sided adhesive tape.

(Configuration 7)

[0182] The optical apparatus according to any one of Configurations 1 to 6, wherein the first sealing member **404** is an adhesive agent.

(Configuration 8)

[0183] The optical apparatus according to any one of Configurations 1 to 7, wherein either one of the first sealing member **404** and the second sealing member **405** also functions as a mask for limiting an optical path.

(Configuration 9)

[0184] The optical apparatus according to any one of Configurations 1 to 8, further includes an electronic display apparatus **29**, wherein the first optical member **401** and the second optical member **402** constitute a magnifying optical system through which the electronic display apparatus **29** is magnified and observed.

(Configuration 10)

[0185] The optical apparatus according to any one of Configurations 1 to 9, wherein an optical effective range of each of the first optical member **401** and the second optical member **402** is shorter in a vertical direction than in a horizontal direction.

(Configuration 11)

[0186] The optical apparatus according to Configuration 9, further includes a diopter adjustment unit **900** for adjusting diopter of the electronic display apparatus **29**, wherein the diopter can be adjusted by the diopter adjustment unit **900** by moving a position of the electronic display apparatus **29** in the optical axis direction.

(Configuration 12)

[0187] The optical apparatus according to Configuration 11, wherein a relative positional relationship between the first optical member **401** and the second optical member **402** does not change when the diopter is adjusted by the diopter adjustment unit **900**.

(Configuration 13)

[0188] The optical apparatus according to any one of Configurations 1 to 12, further includes a third optical member **408** arranged on an inner side of the second optical member **402**, a second holding member **409** for holding the third optical member **408**, and an urging member **410** for urging the first holding member **403** against the second holding member **409**, wherein the second sealing member **405** is in contact with the urging member **410** to compress the urging member **410**, thereby urging the first holding member **403** against the second holding member **409** and determining a position in the optical axis direction.

(Configuration 14)

[0189] The optical apparatus according to any one of Configurations 1 to 13, wherein the abutting portion of the second optical member **402** is a planar face portion **402c** at which the second sealing member **405** is in close contact with the second optical member **402**, and wherein the abutting portion of the first holding member **403** is a planar face portion **403c** at which the second sealing member **405** is in close contact with the first holding member **403**.

[0190] Although the exemplary embodiments of the present disclosure have been described as the above, the invention is not limited to these exemplary embodiments, and many variations and modifications are possible.

[0191] Through the configuration according to the above-described exemplary embodiments, even when either one of the eyepiece lens contact with ambient air and the second-outermost eyepiece lens is made of a plastic material, worsening of surface accuracy, degradation of resolution, and distortion of an image can be reduced.

[0192] Further, through the configuration according to the above-described exemplary embodiments, dew condensation caused by entry of moisture into the inner side of the eyepiece lens can also be suppressed.

[0193] While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0194] This application claims the benefit of Japanese Patent Application No. 2024-026140, filed Feb. 26, 2024, which is hereby incorporated by reference herein in its entirety.

Claims

1. An optical apparatus comprising: a first optical member and a second optical member located on an inner side of the first optical member; a first holding member configured to hold the first optical member and the second optical member; a first sealing member configured to seal a gap between the first optical member and the first holding member; and a second sealing member configured to seal a gap between the first holding member and the second optical member, wherein the second

optical member has an abutting portion on an outer circumferential portion, at which the second sealing member is in contact with the second optical member, wherein the first holding member has an abutting portion on an outer circumferential portion, at which the second sealing member is in contact with the first holding member, wherein the first optical member and the first holding member are sealed with the first sealing member, and wherein a space between the first optical member and the second optical member is sealed.

2. The optical apparatus according to claim 1, wherein the first optical member is a flat plate.
 3. The optical apparatus according to claim 2, wherein a distance between the first optical member and the second optical member on an optical axis is shorter than a thickness of the first optical member.
 4. The optical apparatus according to claim 1, wherein the second optical member is made of plastic.
 5. The optical apparatus according to claim 1, wherein the second optical member is a convex shape lens convexed toward the first optical member.
 6. The optical apparatus according to claim 1, wherein the first sealing member is a double-sided adhesive tape.
 7. The optical apparatus according to claim 1, wherein the first sealing member is an adhesive agent.
 8. The optical apparatus according to claim 1, wherein either one of the first sealing member and the second sealing member limits an optical path.
 9. The optical apparatus according to claim 1, further comprising an electronic display apparatus, wherein the first optical member and the second optical member constitute a magnifying optical system through which the electronic display apparatus is magnified and observed.
 10. The optical apparatus according to claim 9, further comprising a diopter adjustment unit configured to adjust diopter of the electronic display apparatus, wherein the diopter can be adjusted by the diopter adjustment unit by moving a position of the electronic display apparatus in an optical axis direction.
 11. The optical apparatus according to claim 10, wherein a relative positional relationship between the first optical member and the second optical member does not change when the diopter is adjusted by the diopter adjustment unit.
 12. The optical apparatus according to claim 1, wherein an optical effective range of each of the first optical member and the second optical member is shorter in a vertical direction than in a horizontal direction.
 13. The optical apparatus according to claim 1, further comprising: a third optical member arranged on an inner side of the second optical member; a second holding member configured to hold the third optical member; and an urging member configured to urge the first holding member against the second holding member, wherein the second sealing member is in contact with the urging member to compress the urging member, thereby urging the first holding member against the second holding member and determining a position in the optical axis direction.
 14. The optical apparatus according to claim 1, wherein the abutting portion of the second optical member is a planar face portion at which the second sealing member is in close contact with the second optical member, and wherein the abutting portion of the first holding member is a planar face portion at which the second sealing member is in close contact with the first holding member.
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